LIC25 · LAUNCH INTO COMPUTING · PART B

BALANCING INNOVATION AND RESPONSIBILITY IN ENTERPRISE AI

A critical evaluation of one emerging technology - and of Generative AI itself

October 2026 | Anonymous submission | Human research: Slides 2-7 | AI content marked: Slide 8 | Live demo after Slide 8

TECHNICAL OVERVIEW OF AI

02

- Definition: simulation of human intelligence in machines *(Floridi et al., 2018)*
- Subfields: machine learning, deep learning, natural language processing
- Core components: algorithms, data, computing infrastructure
- Current state: large language models, generative AI, multimodal systems
- 84+ national AI ethics guidelines converge on shared principles *(Jobin et al., 2019)*

REAL-WORLD ENTERPRISE APPLICATIONS

03

Healthcare

Medical imaging · drug discovery · predictive diagnostics

Finance

Fraud detection · algorithmic trading · risk management

Manufacturing

Predictive maintenance · quality control

Transportation

Autonomous vehicles · route optimisation

Cybersecurity

Threat detection · anomaly detection

Governance duty

Adopting enterprises assume responsibility alongside benefits (OECD, 2019; NIST, 2023)

ETHICAL CONCERNS IN AI (WITH CASE STUDIES)

04

- Bias and discrimination: data bias, algorithmic bias, disparate impact
- Transparency: black-box problem; GDPR 'right to explanation' *(Goodman and Flaxman, 2017)*
- Fairness is a political-philosophical choice, not just a metric *(Binns, 2018)*
- End-to-end algorithmic auditing closes the accountability gap *(Raji et al., 2020)*

Case study 1 - COMPAS: audit found higher false-positive rates for Black defendants; a statistically accurate system can still discriminate *(Angwin et al., 2016)*

Case study 2 - Amazon recruiting: AI recruiting tool scrapped after downgrading women's CVs learned from male-dominated history *(Dastin, 2018)*

SECURITY RISKS IN AI SYSTEMS

05

- Adversarial attacks: small input perturbations cause misclassification *(Papernot et al., 2016)*
- Model extraction: stealing proprietary models via API queries
- Privacy attacks: neural networks memorise training data *(Carlini et al., 2019)*
- AI-powered threats: enhanced phishing, deepfakes, automated hacking *(Raleigh, 2020)*

Case study 3 - Microsoft Tay: data poisoning in practice - chatbot manipulated into offensive output within 24 hours of deployment *(Vincent, 2016)*

REGULATORY AND GOVERNANCE FRAMEWORKS

- GDPR: constrains automated decision-making; right to explanation *(Goodman and Flaxman, 2017)*
- EU AI Act: risk-based classification shown on the right *(European Commission, 2021)*
- NIST AI RMF: Govern, Map, Measure, Manage *(NIST, 2023)*
- International: OECD AI Principles *(OECD, 2019)*; UNESCO Ethics of AI
- Industry standards: IEEE Ethically Aligned Design; ISO/IEC JTC 1/SC 42
- Remaining gap: regulatory fragmentation across jurisdictions



EU AI Act risk tiers (European Commission, 2021)

HUMAN-AUTHORED RESEARCH FINDINGS

- Evidence-based analysis using peer-reviewed and official sources (full list: Slide 10)
- Ethics needs multi-stakeholder governance, not only technical fixes *(Jobin et al., 2019; Floridi et al., 2018)*
- Security threats evolve continuously, requiring adaptive controls *(Papernot et al., 2016; Raleigh, 2020)*
- Ethical AI extends beyond engineering into organisational responsibility *(Bostrom and Yudkowsky, 2018)*
- Verified, citable sources give claims traceability and credibility

AI-GENERATED CONTENT - PRODUCED WITH A GENERATIVE AI TOOL FOR COMPARISON

AI-GENERATED RESPONSE SUMMARY

- Produced by a Generative AI tool given the identical question (full text: AI_Generated_Response.md)
- **Strengths:** clear structure across all required components; accessible prose; broad coverage
- **Limitations:** no verifiable citations or reference list
- **Limitations:** generalised statements without evidence or case studies
- **Limitations:** no acknowledgement of uncertainty - overconfident generalisation

CRITICAL COMPARATIVE ANALYSIS

Criterion	Human-authored research	AI-generated response
Accuracy & sources	Traceable to peer-reviewed evidence	Unattributed; no verifiable citations
Depth & nuance	Technical specificity; named case studies	Broad overview; general statements
Ethics analysis	Bias types differentiated with mitigation	Concerns mentioned only generally
Professional use	Citable, actionable frameworks	Useful drafting starting point

Key insight: AI excels at structuring and drafting; humans remain accountable for verification and judgement

RECOMMENDATIONS, CONCLUSION & REFERENCES

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Balancing innovation with responsibility:

- Ethics-by-design across the AI lifecycle
- Robust testing: adversarial, bias and performance validation
- Meaningful human oversight for consequential decisions
- Transparency about capabilities, limitations and data use
- Continuous monitoring for drift and emergent behaviours
- Adaptive governance that evolves with understanding

Conclusion: AI augments - does not replace - human expertise and professional judgement

References (Harvard style):

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Vincent, J. (2016) 'Twitter taught Microsoft's AI to be racist...', The Verge, 24 March.